

Calculus II

Area locked by curve

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Outline

1 Areas Locked by Curves

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- 2 Areas in Polar Coordinates

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- The area under a curve $y = F(x)$ from a to b is

$$A = \int_a^b F(x)dx$$

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- How do we know where to put α and β ?
- When $x = a$, t will be either α or β . When $x = b$, t will take the other value.

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Areas in Polar Coordinates

Suppose we have a polar curve $r = f(\theta)$, $a \leq \theta \leq b$.

Definition

We say that the figure obtained as the union of the segments connecting the origin with the points of the curve is the figure *swept* by the curve as θ varies from a to b .

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Theorem

Suppose no two points on the curve lie on the same ray from the origin. Then the area swept by the curve equals $A = \int_a^b \frac{1}{2} (f(\theta))^2 d\theta$.

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Therefore the area swept by the curve **equals the limit of** the sum:

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$$\begin{aligned} A &= \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_{i+1})\sin \Delta}{2} = \lim_{\Delta \rightarrow 0} \frac{\sin \Delta}{\Delta} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_i+\Delta)\Delta}{2} \\ &= \lim_{\Delta \rightarrow 0} \frac{\sin \Delta}{\Delta} \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_i+\Delta)\Delta}{2} \end{aligned}$$

Area swept by a polar curve: justification

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 &= \lim_{\Delta \rightarrow 0} \frac{\sin \Delta}{\Delta} \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_{i+1})\Delta}{2} = ? \cdot \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f(\theta_i)f(\theta_i)\Delta}{2}
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 \text{(Riemann sum)} \quad &= \lim_{\Delta \rightarrow 0} \sum_{i=0}^{N-1} \frac{f^2(\theta_i)\Delta}{2} = \int_a^b \frac{f^2(\theta)}{2} d\theta
 \end{aligned}$$

Example

Find the area enclosed by one loop of the four-leaved rose $r = \cos 2\theta$.

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$$\sin \theta = \frac{1}{2}$$

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$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \frac{\pi}{2} - 2 \cdot 1 + 2 \cdot 0\right) - \left(3 \frac{\pi}{6} - 2 \cdot \frac{1}{2} + 2 \cdot \frac{\sqrt{3}}{2}\right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \\ &= \left(3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot (-1) \right) - \left(3 \frac{\pi}{6} - 2 \cdot 1 + 2 \cdot 1 \right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \cdot \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 1\right) - \left(3 \cdot \frac{\pi}{6} - 2 \cdot \frac{1}{2} + 2 \cdot \frac{\sqrt{3}}{2}\right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \cdot \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0\right) - \left(3 \cdot \frac{\pi}{6} - 2 \cdot 1 + 2 \cdot \frac{\sqrt{3}}{2}\right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0\right) - \left(3 \frac{\pi}{6} - 2 \cdot \frac{1}{2} + 2 \cdot \frac{\sqrt{3}}{2}\right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0\right) - \left(3 \frac{\pi}{6} - 2 + 2\right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= (3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0) - \left(3 \frac{\pi}{6} - 2 + 2 \right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3\frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0\right) - \left(3\frac{\pi}{6} - 2\frac{\sqrt{3}}{2} + 2\right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0\right) - \left(3 \frac{\pi}{6} - 2 \frac{\sqrt{3}}{2} + 2\right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= (3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0) - \left(3 \frac{\pi}{6} - 2 \frac{\sqrt{3}}{2} + 2 \frac{\sqrt{3}}{2} \right) \end{aligned}$$

Example

Find the area that lies within the circle $r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$.

The curves meet if

$$3 \sin \theta = 1 + \sin \theta$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\begin{aligned} A &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 + \sin \theta)^2 d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (9 \sin^2 \theta - (1 + 2 \sin \theta + \sin^2 \theta)) d\theta \\ &= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta \\ &= [3\theta - 2 \sin 2\theta + 2 \cos \theta]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\ &= \left(3 \frac{\pi}{2} - 2 \cdot 0 + 2 \cdot 0\right) - \left(3 \frac{\pi}{6} - 2 \frac{\sqrt{3}}{2} + 2 \frac{\sqrt{3}}{2}\right) \\ &= \pi \end{aligned}$$